Medical Microbiology

Lecture Six

Hypersensitivity

Third stage- College of Dentistry

Dr. Dheyaa Naji Hamza

Dheyaa.naji@alkafeel.edu.iq

Hypersensitivity:

- It refers to exaggerated, harmful, and sometimes life-threatening immune responses produced by an otherwise normal immune system.
- Hypersensitivity reactions are classified into **four types (Type I–IV)** according to the underlying immune mechanism and the time course of the reaction.

1-Hypersensitivity Type I :

- It is also known as **immediate** (Anaphylactic) **hypersensitivity.**
- The reaction may involve skin (eczema), eyes (conjunctivitis), nasopharynx (rhinitis), bronchopulmonary tissues (asthma) and gastrointestinal tract (gastroenteritis).
- The reaction usually takes few minutes from the time of exposure to the antigen (often called allergens).

Allergens: can be complete protein antigens or low–molecular-weight proteins capable of eliciting an IgE response.

Common allergens: Plant pollens, house dust mite, animal products (fur), fungi spores, foods (eggs, milk, and wheat), insect stings (bee), some drugs and chemicals.

Type I Hypersensitivity Mechanism:

- **Main antibody:** IgE
- **Main cells:** Mast cells and basophils

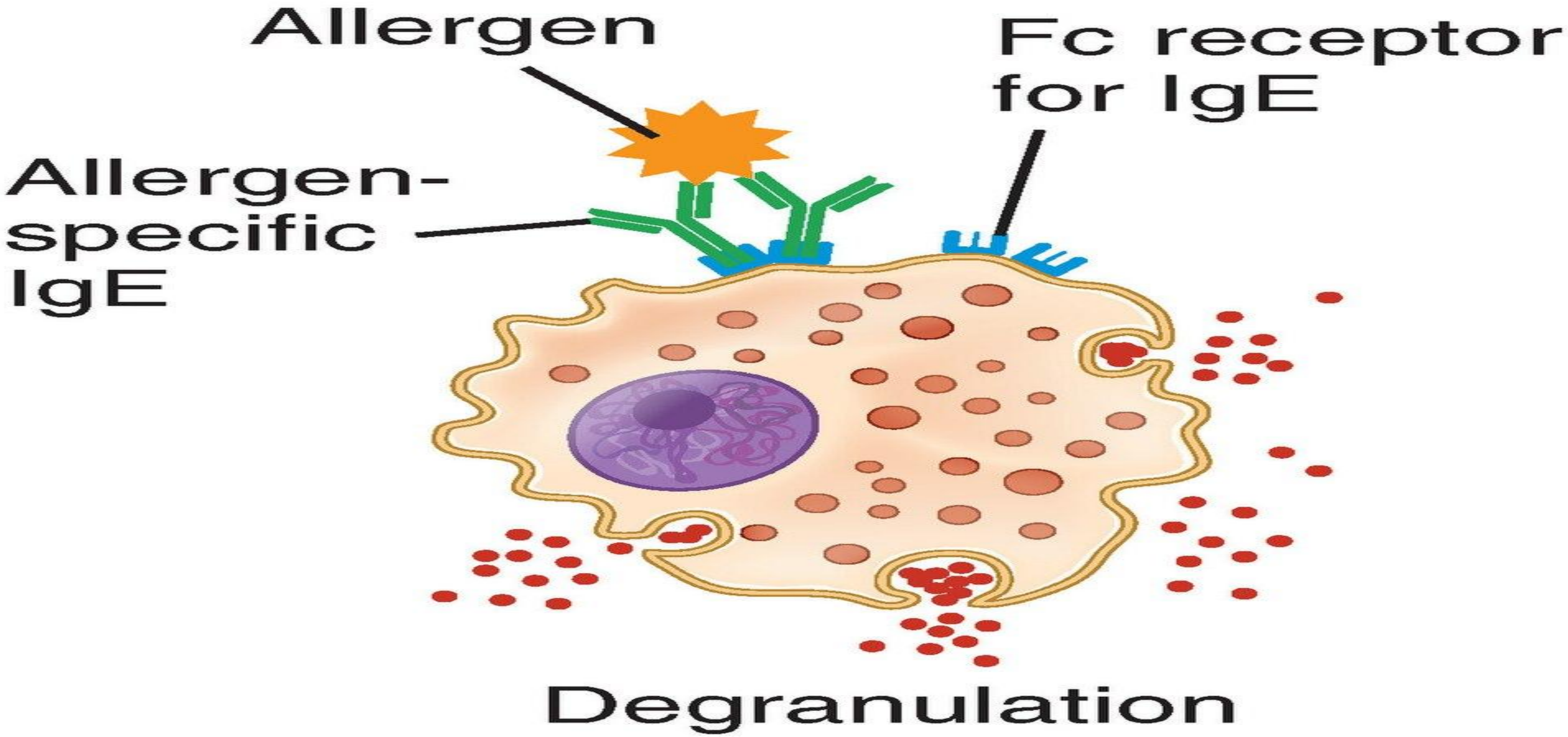
Mechanism:

- First exposure to an allergen leads to IgE production → IgE binds to mast cells → on re-exposure, the allergen cross-links IgE → mast cells release histamine and other mediators → causing allergy symptoms.

On the other words:

- Upon initial exposure to an allergen, IgE antibodies are produced and bind with high affinity to Fc receptors (FcR) on the surface of mast cells and basophils.
- During a subsequent exposure to the same allergen, the allergen cross-links the cell-bound IgE molecules, leading to mast cell degranulation.
- This process releases various pharmacologically active mediators, which induce smooth muscle contraction, increased vascular permeability, and mucus secretion.

- Mast cells can also be activated by stimuli such as exercise, emotional stress, or certain chemicals.
- These responses occur without IgE-allergen interaction and therefore are not classified as hypersensitivity reactions.
- However, they produce similar clinical manifestations and are referred to as anaphylactoid reactions.


Allergen
Fc receptor for IgE
Allergen-specific IgE
Degranulation

Mediators of Immediate (Type I) Hypersensitivity:

A. Preformed mediators (stored in mast cell granules):

- Histamine
- Eosinophil chemotactic factor (ECF)
- Serotonin
- Kininogenase

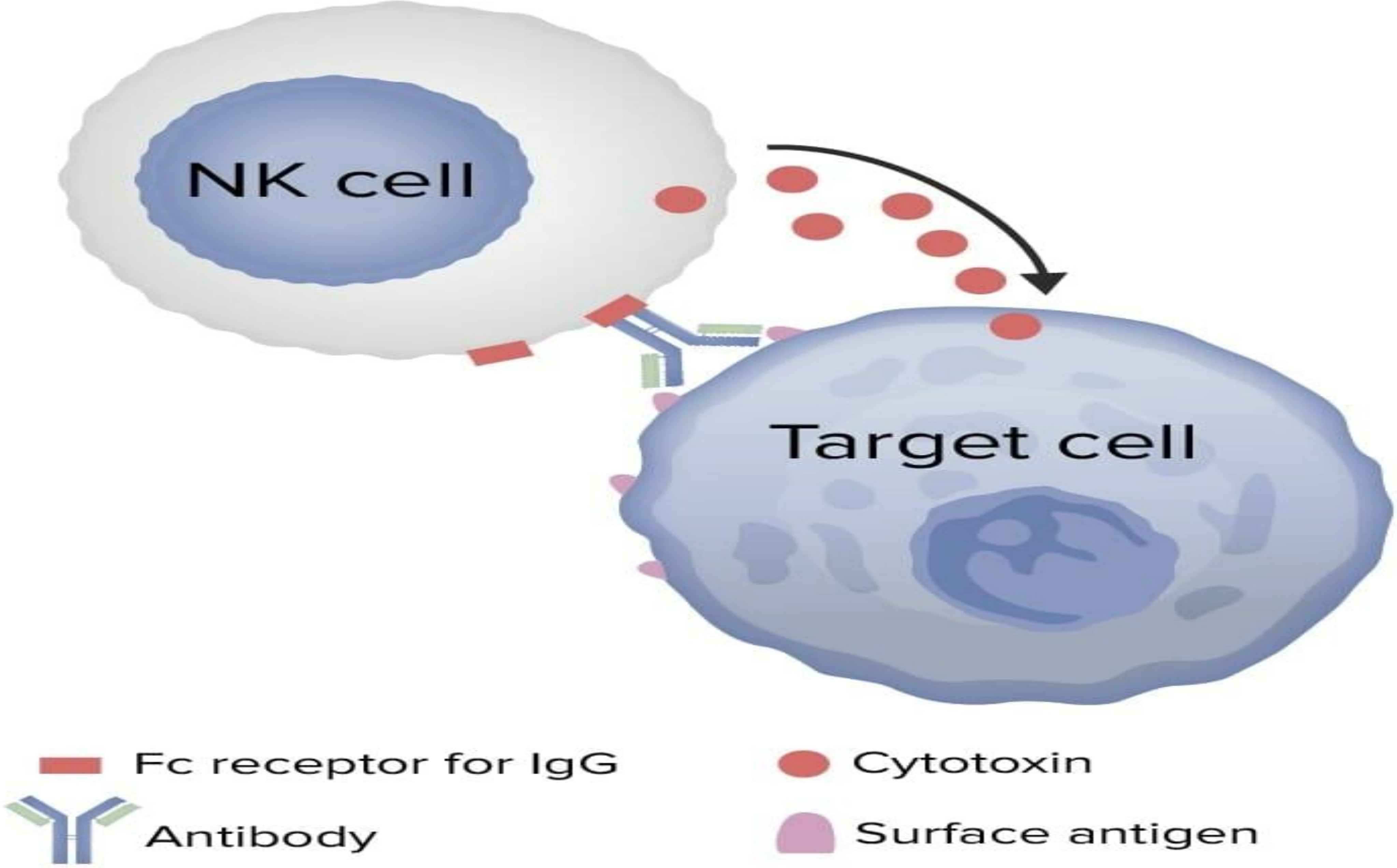
B. Newly synthesized mediators (formed after activation):

- Leukotrienes
- Prostaglandins
- Platelet-activating factor (PAF)

2-Type II – Cytotoxic Hypersensitivity:

Main antibodies: IgG or IgM

- **Mechanism:**
  Antibodies bind to antigens on the surface of cells → activate complement or NK cells → cause cell destruction (lysis or phagocytosis).
- **Time:** Minutes to hours
- **Examples:** Hemolytic anemia, transfusion reactions, Rh incompatibility


NK cell
Target cell
Fc receptor for IgG
Cytotoxin
Antibody
Surface antigen

Some Clinical Features of Type II Hypersensitivity:

1-Hemolytic disease of the newborn (Erythroblastosis fetalis):
Occurs when an Rh-negative mother carries an Rh-positive fetus. Fetal blood entering maternal circulation sensitizes the mother, leading to the production of anti-Rh IgG antibodies that cross the placenta and destroy fetal red blood cells in subsequent pregnancies.

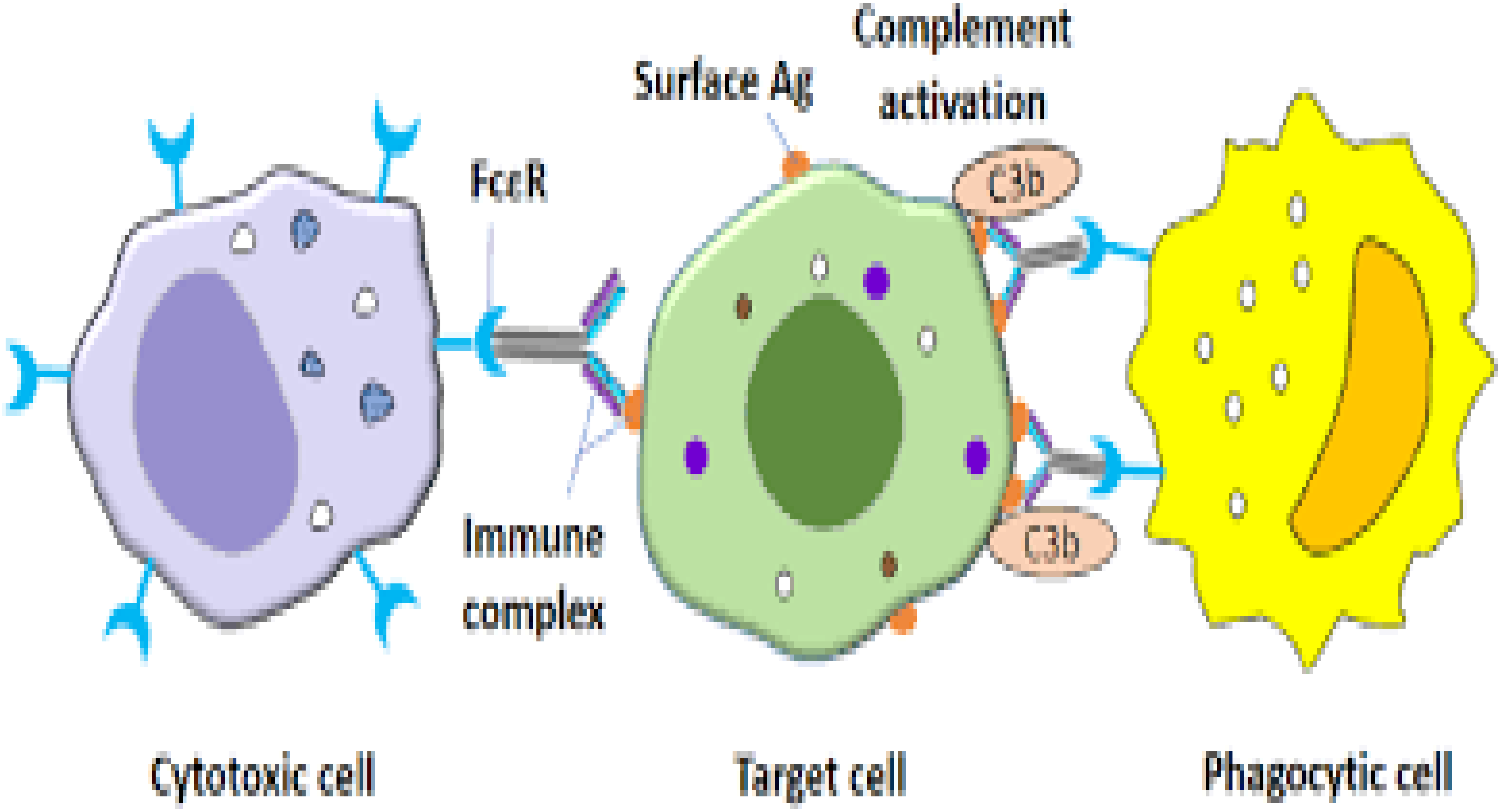
2-Transfusion reactions:
Caused by antibodies reacting with incompatible donor red blood cells, leading to cell lysis.

3-Leukocyte (white blood cell) destruction:
Occurs when antibodies target white blood cells, resulting in their lysis.

4-Goodpasture's syndrome:
Autoantibodies attack the basement membrane of the lungs and kidneys, causing bleeding and organ damage.

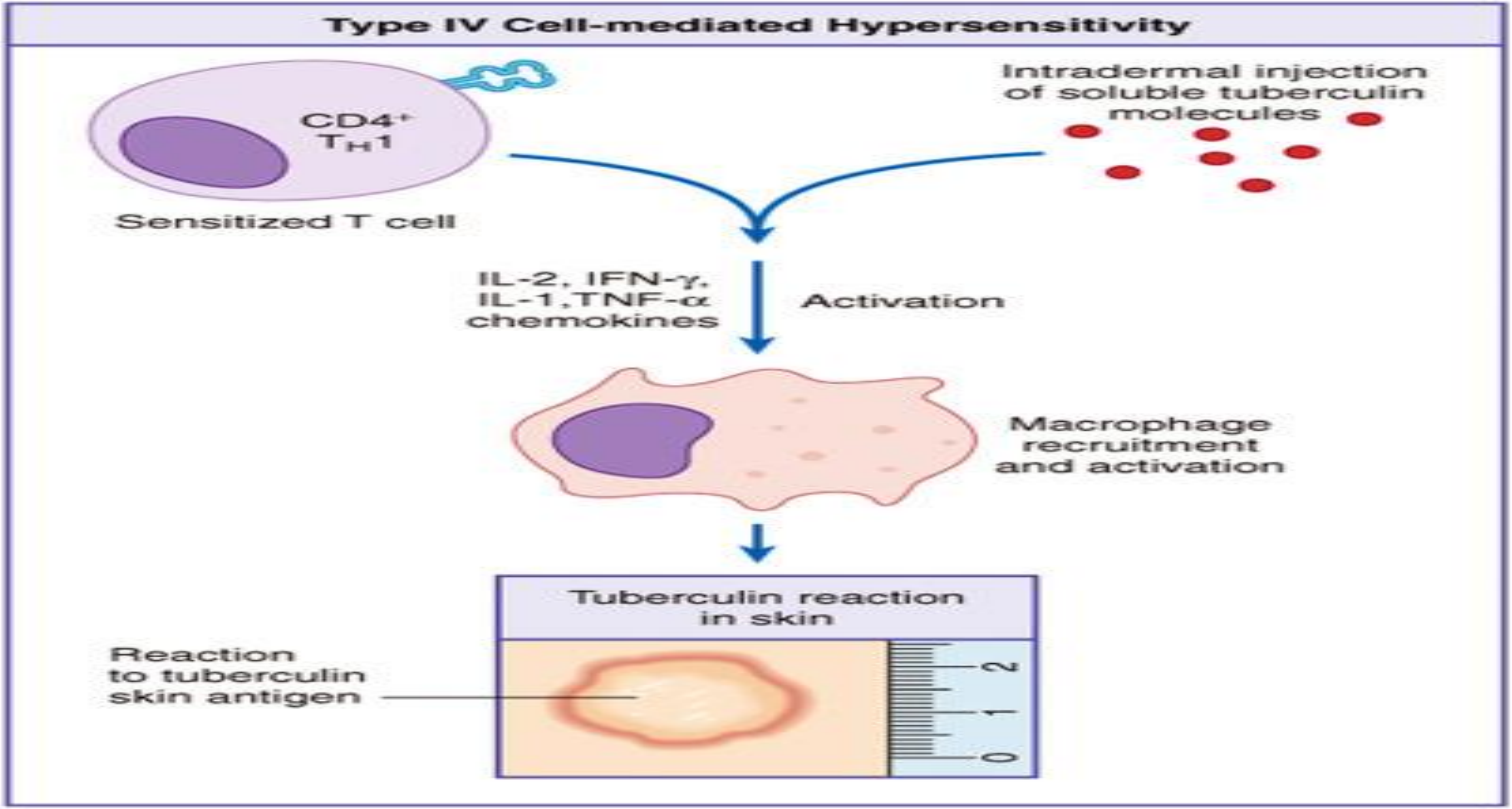
3-Type III – Immune Complex-Mediated Hypersensitivity:

- **Main antibodies:** IgG and IgM
- **Mechanism:** Antigen–antibody complexes form in circulation → deposit in tissues → activate complement → cause inflammation and tissue damage.
- **Time:** 3–8 hours.
- **Examples:** Serum sickness, systemic lupus erythematosus (SLE).


Complement activation
Surface Ag
FcεR
C3b
C3b
Immune complex
Cytotoxic cell
Target cell
Phagocytic cell

4-Type IV – Delayed (Cell-Mediated) Hypersensitivity:

- **Mediated by:** T lymphocytes (not antibodies)
- **Mechanism:**
  Sensitized T cells recognize antigen → release cytokines → attract macrophages and other immune cells → cause tissue injury.
- **Time:** 24–72 hours after exposure
- **Examples:** Tuberculin skin test, contact dermatitis, graft rejection.


Type IV Cell-mediated Hypersensitivity
CD4+ TH1
Sensitized T cell
Intradermal injection of soluble tuberculin molecules
IL-2, IFN-γ, IL-1, TNF-α chemokines
Activation
Macrophage recruitment and activation
Tuberculin reaction in skin
Reaction to tuberculin skin antigen
2
1
0

Type	Mediator	Onset Time	Mechanism	Examples
I	IgE	Seconds–minutes	Mast cell degranulation → histamine release	Asthma, anaphylaxis
II	IgG / IgM	Minutes–hours	Antibody binds to cell surface → cell lysis	Hemolytic anemia
III	IgG / IgM	3–8 hours	Immune complex deposition → inflammation	SLE, serum sickness
IV	T cells	24–72 hours	T-cell–mediated inflammation	Tuberculin test, contact dermatitis

Any Question?

Thank you